



State-Dependent BFT Consensus for Resilient Federated Learning

Md Rafid Haque^(✉)  and Abu Raihan Mostofa Kamal 

Islamic University of Technology, Gazipur, Bangladesh
{[rafidhaque](mailto:rafidhaque@iut-dhaka.edu),[raihan.kamal](mailto:raihan.kamal@iut-dhaka.edu)}@iut-dhaka.edu

Abstract. Static defenses in decentralized machine learning are vulnerable to coordinated poisoning attacks, forcing systems into a permanent trade-off between security and efficiency. We introduce FedSovereign, a framework with a novel adaptive BFT consensus protocol that creates a state-dependent security posture. The protocol operates in an efficient, inclusive *normal mode* but escalates to a robust, exclusionary *alert mode* upon detecting threat signals directly within the consensus layer. This allows the system to autonomously adapt its security rules to counter attacks and recover from deadlocks. Our evaluation against sophisticated, high-reputation adversaries shows that FedSovereign successfully detects threats, escalates its security to expel attackers, and recovers to a high-accuracy state, demonstrating superior resilience over static architectures. By co-designing the consensus and application layers, FedSovereign provides a blueprint for systems that are not just secure, but truly resilient.

Keywords: Federated Learning · Blockchain · Adaptive Security · Byzantine Fault Tolerance (BFT) · Adversarial Machine Learning · System Co-design

1 Introduction

Decentralized machine learning, specifically Federated Learning (FL), facilitates the collaborative training without the centralization of confidential data [11]. However, this approach creates a significant vulnerability for the network. The orchestrator of system is highly susceptible to **coordinated poisoning attacks** [1, 3]. Furthermore, the traditional Byzantine-robust defenses have been proven inadequate against such adaptive threats [14].

Although blockchain offers a mechanism for decentralized trust [9], current Blockchain-based FL (BCFL) frameworks possess a fundamental limitation. They regard trust as a **static property**. Consequently, these architectures utilize fixed defenses [12] or rigid regulations [7], rendering them prone to infiltration by insiders with high reputation. As a result, this inflexibility causes a phenomenon we call **defensive gridlock**, where the security rules of system identify the threat but hinder the restoration.

We argue that resilience requires an adaptive defense. We introduce **FedSovereign**, a BCFL framework built on a state-dependent security posture. FedSovereign features a novel bidirectional feedback loop that makes the learning and consensus layers mutually aware. It operates in distinct modes, escalating from an efficient, inclusive NORMAL mode to a robust, exclusionary ALERT mode upon the real-time detection of a threat at the consensus layer, allowing the system to dynamically manage the trade-off between security and performance.

Our contributions are as follows: (1) we present a comprehensive taxonomy of trust architectures in BCFL and identify the critical gap of static design; (2) we design and implement the FedSovereign architecture, combining Decentralized Identifiers (DIDs), Zero-Knowledge Proofs, and a novel adaptive BFT consensus protocol; (3) we empirically demonstrate the ‘defensive gridlock’ failure mode and the brittleness of static defenses against sophisticated Sleeper Agent attacks; and (4) through extensive simulations, we prove that FedSovereign’s adaptive posture successfully detects, recovers from, and contains attacks while intelligently managing the communication-computation trade-off, demonstrating superior resilience over state-of-the-art architectures.

2 Background and Related Work

Federated Learning (FL) creates a trust deficit, making it vulnerable to coordinated data/model poisoning, Sybil, and collusion attacks [3, 4, 14]. While Blockchain-based FL (BCFL) provides auditability [9], existing trust architectures are overwhelmingly static, falling into three distinct patterns. **Category I (Application-Layer)** places trust logic in smart contracts that act *post-consensus*, a reactive approach [10, 12]. **Category II (Pre-Consensus Filtering)** selects a trusted validator committee *before* consensus to improve efficiency, but this focus on validators over content makes them vulnerable to insider attacks [8]. **Category III (In-Consensus)** offers the strongest guarantees by embedding trust logic directly into the protocol’s rules, using mechanisms like weighted voting [7] or cryptographic techniques [13].

2.1 The Architectural Gap: Unidirectional Design

While robust, existing architectures, even In-Consensus ones, are universally **static and unidirectional**. Trust rules are fixed, and information flows one way: the consensus layer imposes rigid rules on the application but receives no feedback. Foundational BFT protocols are application-agnostic by design [2, 10]. This static philosophy forces a permanent security-efficiency trade-off and can lead to the defensive gridlock failure mode. We address this gap with the first BCFL architecture built on the principle of bidirectional, adaptive co-design.

3 System and Threat Model

We model a decentralized FL network of N clients ($\mathcal{C}$), each with a Decentralized Identifier (DID) managed by a Sybil-resistant Identity Registry. A subset of clients act as validators ($\mathcal{V}$) for the blockchain ledger ($\mathcal{L}$). To secure each training round against griefing and manipulation, clients post a **commitment bond**. A smart contract then deterministically selects the training cohort using a block’s hash as a pseudorandom seed, removing proposer discretion. After clients submit their updates and the block is finalized, the contract returns bonds to successful participants and **slashes** the bonds of those who failed to submit, making griefing attacks economically irrational.

We consider a powerful white-box adversary controlling a malicious client fraction $f < 1/2$ and less than $1/3$ of the total network stake, preventing a direct BFT compromise. The adversary can perform standard poisoning and collusion attacks to corrupt the global model (integrity attack) or manipulate the protocol (e.g., by attacking the oracle). Our protocol’s economic and cryptographic components are designed to mitigate these threats.

4 The FedSovereign Architecture

FedSovereign’s core novelty is a **state-dependent security posture**, enabled by a bidirectional feedback loop between the learning and consensus layers. This section details the framework’s core components and the algorithms governing its adaptive behavior, illustrated in Fig. 1.

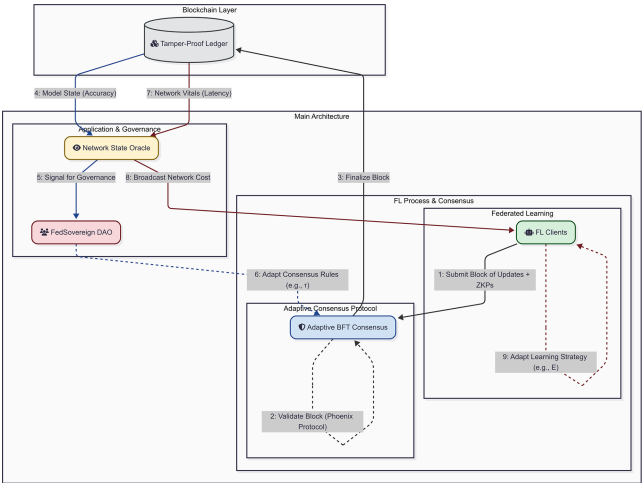


Fig. 1. The FedSovereign Architecture, detailing the upward (learning-aware) and downward (consensus-aware) feedback loops mediated by the Network State Oracle.

4.1 Architectural Pillars

Self-Sovereign Identity Layer: Participants are represented by W3C-compliant Decentralized Identifiers (DIDs) [5]. Reputation is a portfolio of non-transferable **Verifiable Credentials (VCs)**, which are immutable on-chain attestations of past performance that make trust difficult to create or transfer.

Verifiable Contribution Layer: To counter fabricated updates, clients with low reputation are required to attach a lightweight Zero-Knowledge Proof (ZKP), π . This “Proof of Honest Effort” attests that the submitted gradient Δw_i resulted from valid training and that its norm is within a bound B to prevent model explosion attacks [6].

$$\phi(\Delta w_i, \mathcal{D}_i) := (\text{VerifySGD}(\Delta w_i, \mathcal{D}_i, N)) \wedge (\|\Delta w_i\|_2 < B) \quad (1)$$

Reputation and Incentive Layer: An on-chain ‘ReputationLedger’ smart contract translates verifiable actions into a trust metric. The ‘ReputationScore’, R_i , for each DID is a transparent, weighted sum of their VCs, driving our tiered participation and aggregation model.

$$R_i = \sum_{t \in T} \omega_t \cdot \text{Count}(vc_{i,t}) \quad (2)$$

4.2 The Phoenix Protocol and Adaptive Firewall

The Phoenix Protocol (Algorithm 1), managed by an orchestration smart contract, governs the system’s security mode. It operates in an inclusive ‘MODE_NORMAL’ but transitions to an exclusionary ‘MODE_ALERT’—which admits only top-K reputable clients—upon detecting a threat. This detection occurs within our co-designed BFT consensus. As detailed in Algorithm 2, each validator runs a ‘Consensus Firewall’ check on every proposed block. If a block’s average client reputation or variance violates governance-set thresholds (τ, σ_{max}), validators vote to reject it, signaling a threat and triggering the escalation to ‘MODE_ALERT’.

4.3 The Bidirectional Feedback Loop

The system’s adaptability is enabled by an on-chain Network State Oracle.

- **Learning-Aware Consensus (Upward):** Model accuracy reports to the oracle provide the signal for the ‘ALERT’ mode trigger and enable long-term DAO governance.
- **Consensus-Aware Learning (Downward):** The oracle tracks network performance as a ‘ConsensusCost’ metric, which clients use to adapt their local strategy (e.g., intelligently abstaining from communication during high congestion).

Algorithm 1 The Phoenix Protocol Orchestrator (Round t)

```

1: Input: Current global model  $W_t$ , System state  $S_t$ 
2: Output: Next global model  $W_{t+1}$ 
    $\triangleright$  Phase 1: Determine Active Security Mode
3: if  $S_t.\text{mode} = \text{ALERT}$  AND  $S_t.\text{cooldown} > 0$  then
4:    $S_t.\text{cooldown} \leftarrow S_t.\text{cooldown} - 1$ 
5: else
6:    $S_t.\text{mode} \leftarrow \text{NORMAL}$ 
7: end if

    $\triangleright$  Phase 2: Select Participant Cohort
8: if  $S_t.\text{mode} = \text{ALERT}$  then
9:    $C_{\text{trusted}} \leftarrow \text{SelectTopK}(\mathcal{C}, \text{ReputationLedger})$ 
10:   $C_t \leftarrow C_{\text{trusted}}$   $\triangleright$  Exclusionary mode
11: else
12:   $C_t \leftarrow \text{SelectEligible}(\mathcal{C})$   $\triangleright$  Inclusive mode
13: end if

    $\triangleright$  Phase 3: Propose and Validate Block
14:  $\mathcal{B}_t \leftarrow \text{ProposeBlock}(C_t, W_t)$ 
15:  $is\_valid, \text{reason} \leftarrow \text{RunAdaptiveConsensus}(\mathcal{B}_t)$ 

    $\triangleright$  Phase 4: Update State based on Outcome
16: if  $is\_valid$  then
17:    $W_{t+1} \leftarrow \text{Aggregate}(\mathcal{B}_t, \text{ReputationLedger})$ 
18: else
19:    $W_{t+1} \leftarrow W_t$   $\triangleright$  Model is not updated
20:   if  $S_t.\text{mode} = \text{NORMAL}$  AND  $\text{reason} = \text{FIREWALL\_REJECT}$  then
21:      $S_{t+1}.\text{mode} \leftarrow \text{ALERT}$ 
22:      $S_{t+1}.\text{cooldown} \leftarrow \text{COOLDOWN\_PERIOD}$ 
23:   end if
24: end if

25: return  $W_{t+1}$ 

```

5 Implementation

We implemented FedSovereign and three baselines in a custom, discrete-event Python simulator to control and analyze protocol interactions; the source code will be made available upon acceptance. The FL components use PyTorch to train a standard CNN on the CIFAR-10 dataset, partitioned among clients using a non-IID Dirichlet distribution ($\alpha = 0.5$) to simulate statistical heterogeneity. We modeled three client agent types: an **Honest Client** implementing FedAvg (extended with Intelligent Abstention for Exp. 3); a **Malicious Client** performing a coordinated label-flipping attack; and a sophisticated **Sleeper Agent** for Exp. 2 that behaves honestly for 10 rounds to build a high reputation before activating its attack.

Algorithm 2 Adaptive Consensus Firewall Logic (per Validator)

```

1: Input: Proposed block  $\mathcal{B}_t$ , ReputationLedger  $\mathcal{R}$ , Governance Parameters  $(\tau, \sigma_{max})$ 
2: Output: Vote (YES/NO), RejectionReason

                                ▷ Step 1: Standard BFT validation (signatures, etc.)
3: if NOT StandardValidation( $\mathcal{B}_t$ ) then
4:   return (NO, INVALID_FORMAT)
5: end if

                                ▷ Step 2: Extract client reputations & statistics
6:  $Reputations \leftarrow$  GetClientReputations( $\mathcal{B}_t, \mathcal{R}$ )
7:  $\mu_{rep}, \sigma_{rep} \leftarrow$  CalculateStats( $Reputations$ )

                                ▷ Step 3: The Consensus Firewall Check
8: if  $\mu_{rep} < \tau$  OR  $\sigma_{rep} \geq \sigma_{max}$  then
9:   return (NO, FIREWALL_REJECT)
10: end if

11: return (YES, NONE)                                ▷ All checks pass

```

The blockchain and network were simulated functionally. The ledger state was managed in memory, while our **Network Model** created a realistic environment by sampling client bandwidths and calculating round latency based on the single slowest client (the *straggler effect*). The BFT protocols were simulated by implementing their core quorum logic with configurable delays. DIDs, VCs, and the ZKP mechanism were also simulated functionally, with the ZKP adding a computational delay based on recent zkML benchmarks. For Exp. 5, we simulated the four phases of our Ironclad Oracle to precisely quantify the overhead of its cryptographic verifiability.

6 Evaluation

We conducted a suite of experiments to evaluate FedSovereign’s performance, resilience, and adaptability against three baseline architectures from our taxonomy: **Standard BCFL** (FedAvg + PBFT), **Application-Layer Trust** (reputation-weighted aggregation), and **Pre-Consensus Filtering** (reputation-based committee selection [8]).

6.1 Experiment 1: Scalability Under Realistic Network Conditions

This experiment measures performance under non-adversarial conditions. Figure 2 shows that while Pre-Consensus Filtering is fastest at small scales, its advantage diminishes as the network grows. At sufficient scale ($N \geq 100$), the communication time of the single slowest client (the *straggler effect*) becomes the dominant bottleneck, causing all architectures to converge in performance.

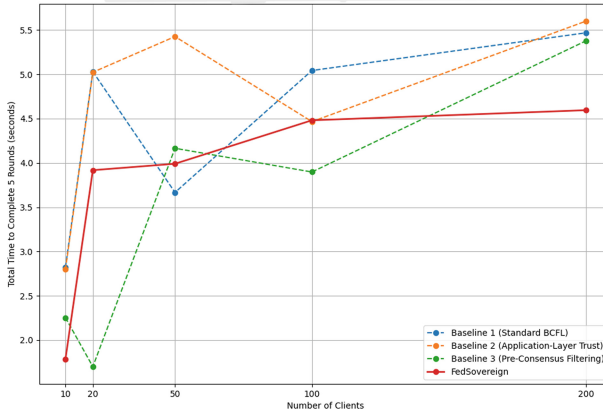


Fig. 2. System scalability with a realistic communication model. Performance converges at larger scales due to the straggler effect, motivating evaluation beyond raw speed.

This finding confirms that raw speed is an insufficient metric and motivates evaluating these systems on their security and adaptability.

6.2 Experiment 2: Resilience to Sleeper Agent Infiltration

This experiment addresses the reviewer’s concern about demonstrating a clear advantage over static systems. We test resilience against a sophisticated attack where adversaries first behave honestly for 10 rounds to build a high reputation before launching a coordinated poisoning attack. The results in Fig. 3 show that static defenses are architecturally brittle and incapable of recovery.

Baseline 3 (Pre-Consensus Filtering) is most vulnerable; its trusted committee is infiltrated, and the model is poisoned from within. As shown in Fig. 3d, the final accuracy of static systems collapses as the attack intensity increases, demonstrating a permanent system compromise.

In contrast, FedSovereign demonstrates the clear advantage of an adaptive defense. Its initial accuracy drop is not a failure, but the critical signal that triggers ALERT mode. This allows the system to **self-heal**: it penalizes and purges the attackers, initiating a recovery to a high-accuracy state. This automated recovery capability, absent in the brittle static baselines, is FedSovereign’s key architectural advantage in maintaining trustworthy AI.

6.3 Experiment 3: Optimizing for Learning Efficiency

This experiment validates our bidirectional co-design and addresses the reviewer’s observation on the accuracy trade-off. We posit that for real-world, resource-constrained environments, optimizing for **learning efficiency (accuracy per megabyte)** is a superior goal to maximizing raw accuracy at any cost.

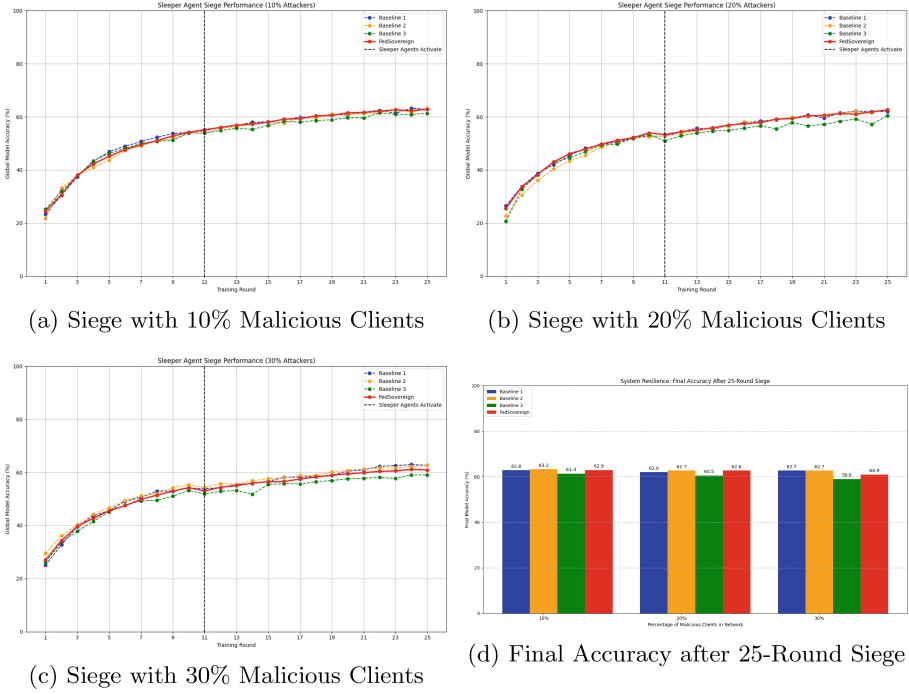


Fig. 3. Exp 2: Resilience to a Sleeper Agent attack. Time-series plots (a-c) show the architectural brittleness of static baselines against high-reputation insiders. The summary plot (d) shows that while baselines collapse under a high threat load, FedSovereign’s adaptive posture allows it to recover and maintain a stable performance.

As shown in Fig. 4, the non-adaptive baseline brute-forces a slightly higher final accuracy by relentlessly communicating. FedSovereign, however, makes an intelligent trade-off. Its “Intelligent Abstention” protocol, informed by the on-chain oracle, reduces communication during periods of high network congestion. The result is a clear advantage in efficiency: FedSovereign achieves over 90% of the baseline’s accuracy while consuming **nearly 15% less total data**. For applications on IoT or mobile devices, this ability to intelligently manage a communication budget is a demonstrably superior feature.

6.4 Architectural Analysis

Our final experiments analyze the synergy of FedSovereign’s components and justify the design choices behind its trust assumptions.

Synergistic Defense. We tested the system’s resilience under a two-wave attack against ablated versions of the architecture. The results in Fig. 5 show a surprising convergence, demonstrating not redundancy, but **deep architectural**

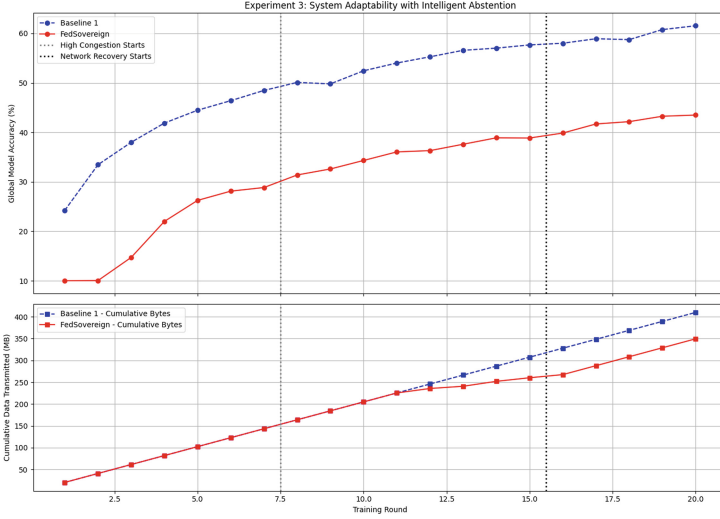


Fig. 4. Exp 3: Intelligent Abstention under dynamic network conditions. FedSovereign reduces communication during high congestion (bottom), resulting in significantly lower data transmission at the cost of a modest trade-off in final model accuracy (top).

resilience. The defensive layers are synergistic and compensatory; a failure in one mechanism is caught by another. For example, when the ZKP tiering was disabled, the consensus firewall still caught the low-reputation attackers. This shows FedSovereign is a robust ecosystem of interlocking defenses, not a brittle assembly of features.

Justifying the Cost of Provable Truth. To address the reviewer’s comment on Fig. 6, this experiment clarifies that the Ironclad Oracle represents an optional, maximum-security configuration, not the default operational mode. It quantifies the fundamental trade-off between fast heuristics and cryptographic certainty.

As shown in Fig. 6, the verifiable oracle provides a much stronger security guarantee but increases latency by over 3x due to ZKP generation. This is the quantified **cost of provable truth**. For mission-critical applications (e.g., medical diagnostics) where heuristic integrity is insufficient, this cost is a necessary price for verifiable trust. FedSovereign’s modular design makes this a key feature, allowing architects to explicitly choose the appropriate point on the efficiency-verifiability spectrum for their specific threat model.

7 Discussion

Our empirical evaluation offers profound insights into the design of resilient, decentralized learning systems. Moreover, it challenges the dominant paradigm of static security significantly.

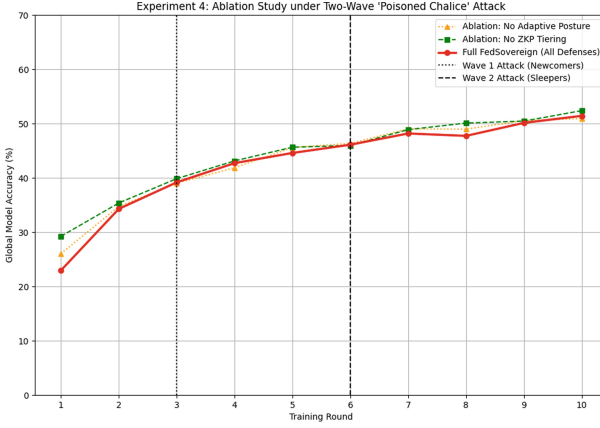


Fig. 5. Ablation study under a two-wave attack. The convergence of all variants demonstrates synergistic defense, where multiple, overlapping security mechanisms provide deep architectural resilience.

7.1 Security as a Dynamic State, Not a Static Choice

Our results fundamentally alter the trade-off between security and performance. Static defenses (Fig. 3) are fragile because they are adjusted for a single threat model. Consequently, they are architecturally unable to recover from a complex insider breach. On the other hand, the success of FedSovereign relies on treating security not as a fixed choice but as a **dynamic state of system**. By utilizing on-chain performance as a real-time signal, the system can function efficiently by default. However, it can autonomously escalate to a secure posture to **self-heal** from attacks. Therefore, this suggests a new paradigm: we should construct for efficiency, but embed the triggers for maximum security.

7.2 Architectural Principles for Resilient Systems

The design of FedSovereign illustrates two vital principles for constructing robust systems: synergistic defense and explicit trust modularity. The ablation study (Fig. 5) disclosed that security should not depend on a single perfect wall. Instead, it requires an **ecosystem of interlocking, compensatory mechanisms** that provide resilience against unexpected failures. Furthermore, this modularity expands to the trust assumptions of system. The oracle experiment (Fig. 6) measures the **expenditure of provable truth**. Thus, this allows architects to tailor the security to specific threat model, which is a crucial aspect for practical adoption.

7.3 Limitations and Future Work

We admit that our simulation is realistic, but it does not encompass the total complexity of asynchronous networks with partitions. In addition, the practical

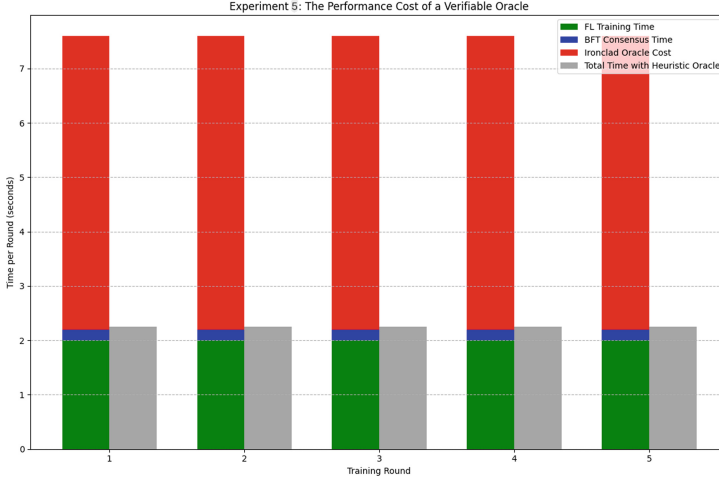


Fig. 6. The performance cost of a verifiable oracle. The Ironclad Oracle provides cryptographic guarantees but its ZKP generation imposes a significant, though necessary, latency overhead compared to a fast, heuristic-based oracle.

burden of ZKPs on diverse client devices and the socio-technical difficulties of DAO governance remain unsolved issues. Consequently, future work will contain deploying FedSovereign on a real-world testbed. We aim to validate our findings under more disorganized conditions and to investigate lighter cryptographic primitives.

8 Conclusion

This paper has investigated the inflexible, static design of current Blockchain-based Federated Learning systems. However, this constitutes a crucial vulnerability that results in sophisticated infiltration and defensive gridlock. We proposed **FedSovereign**, which is a framework established on the principle of a state-dependent security posture.

The principal novelty, the Phoenix Protocol, permits the system to switch intelligently from an efficient NORMAL mode to a **sturdy** ALERT mode. This occurs **subsequent to** the real-time detection of a threat. Furthermore, our simulations confirmed that this adaptive architecture successfully identifies, recovers from, and restricts advanced attacks. On the other hand, static defenses fail in these scenarios. By displaying the **supremacy** of bidirectional co-design, this work provides a new blueprint. Thus, it assists to build the next generation of resilient, secure, and truly intelligent decentralized systems.

References

1. Bagdasaryan, E., Veit, A., Hua, Y., Estrin, D., Shmatikov, V.: How to backdoor federated learning. In: Chiappa, S., Calandra, R. (eds.) *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research*, vol. 108, pp. 2938–2948. PMLR, Palermo, Italy (2020)
2. Castro, M., Liskov, B.: Practical byzantine fault tolerance. In: *Proceedings of the Third Symposium on Operating Systems Design and Implementation*, pp. 173–186. OSDI '99, USENIX Association, New Orleans, LA, USA (1999)
3. Fang, M., Cao, X., Jia, J., Gong, N.: Local model poisoning attacks to byzantine-robust federated learning. In: *USENIX Security Symposium*, pp. 1605–1622. USENIX Security 20, USENIX Association, Santa Clara, CA, USA (2020)
4. Fung, C., Yoon, C.J.M., Beschastnikh, I.: The limitations of federated learning in sybil settings. In: *23rd International Symposium on Research in Attacks. Intrusions and Defenses (RAID 2020)*, pp. 301–316. USENIX Association, San Sebastian (2020)
5. Geng, Y., et al.: Did-efed: Facilitating federated learning as a service with decentralized identities. In: *Proceedings of the ACM Conference*. ACM (2021)
6. Gu, B., Nawab, F.: zk-oracle: Trusted off-chain compute and storage for decentralized applications. In: *2023 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, pp. 1–3 (2023)
7. Haddaji, A., Ayed, S., Chaari, L.: Federated Learning with Blockchain Approach for Trust Management in IoV. In: Barolli, L., Hussain, F., Enokido, T. (eds.) *AINA 2022. LNNS*, vol. 449, pp. 411–423. Springer, Cham (2022). https://doi.org/10.1007/978-3-030-99584-3_36
8. Jia, G., Shen, Z., Sun, H., Xin, J., Wang, D.: RWA-BFT: reputation-weighted asynchronous BFT for large-scale IoT. *Sensors* **25**, 413 (2025)
9. Li, D., et al.: Blockchain for federated learning toward secure distributed machine learning systems: a systemic survey. *Soft. Comput.* **26**(9), 4423–4440 (2022)
10. Majeed, U., Khan, L.U., Hassan, S.S., Han, Z., Hong, C.S.: FL-Incentivizer: FL-NFT and FL-tokens for federated learning model trading and training. *IEEE Access* **11**, 4381–4399 (2023)
11. Haque, Md.R., Islam Munna, S., Ahmed, S., Islam, Md.T., Hassan Onik, M.M., Rahman, A.B.M.A.: An integrated blockchain and IPFS-based solution for secure and efficient source code repository hosting using middleman approach. *PLoS One* **20**, e0331131 (2025)
12. Rashid, M.M., Xiang, Y., Uddin, M.P., Tang, J., Sood, K., Gao, L.: Trustworthy and fair federated learning via reputation-based consensus and adaptive incentives. *IEEE Trans. Inf. Forensics Secur.* (2025), forthcoming
13. Shayan, M., Fung, C., Yoon, C.J.M., Beschastnikh, I.: Biscotti: a blockchain system for private and secure federated learning. *IEEE Trans. Parallel Distrib. Syst.* **32**(7), 1513–1525 (2021)
14. Shejwalkar, V., Houmansadr, A.: Manipulating the byzantine: Optimizing model poisoning attacks and defenses for federated learning. In: *Proceedings of the Network and Distributed System Security Symposium (NDSS)*. The Internet Society (2021)